

Organic Compounds

Compounds containing carbon and hydrogen only or their derivatives are called organic compounds.

Types:

There are three types of organic compounds.

Acidic organic compounds

Basic organic compounds

Neutral organic compounds

1- Acidic organic compounds:

Organic compounds which give acidic solution or donate hydrogen ion are called acidic compounds.
e.g: -COOH , Phenol etc.

2- Basic organic compounds:

Organic compounds which give basic solution or accept hydrogen ions are called basic compounds.
e.g: Amines, Amides etc.

3- Neutral organic compounds:

Organic compounds which are not acidic or basic are called neutral compounds. e.g: Glucose etc.

Separation of a mixture:

There are two methods for separation of a mixture.

Physical separation method

Chemical separation method

1- Physical Separation Method:

It depends upon the solubility of one or more of compounds of a mixture in any one of the solvents usually applied, that are water, ether, chloroform, benzene, CCl_4 , petroleum ether etc.

- Try to separate or separation of a mixture with the small quantity at first if the result is positive then add a quantity of solvent to approximately three grams of the mixture.
- Try to separate or separation of a mixture with the small quantity at first
- Filter and wash the residue with some solvent to avoid contamination.
- Recover the soluble compounds by evaporation of the filtration on a water bath.
- Test the residue and recovered components by usual methods for identification.

2- Chemical Separation Method:

If the given mixture is totally soluble or insoluble in physical separation method by the solvent then apply chemical separation method.

The acidic, basic or neutral compounds of the mixture is identified with litmus paper.

1(a) If the mixture is acidic in nature and evolves CO_2 with standard solution of NaHCO_3 then treat about 3g of the mixture with standard NaHCO_3 solution till it (will) gives no effervescence. Separate the residue by the process of filtration. Acidify the filtration with conc. HCl and filter the precipitated acid. Identify the residue and recovered acid after drying.

1(b) If the mixture is acidic but does not evolve CO_2 with standard solution of NaHCO_3 then treat 3g of the mixture with 10% NaOH solution and filter. Dry residue and recover the acid from filtration by precipitated with con. HCl .
Identify the residue and recovered components by means of usual methods.

2- If the mixture is basic in nature then treat 3g of the given mixture with conc. HCl warm and filter by treating the dry residue for identification.
Recover the second component from filtration by treating it with 10% NaOH till alkaline.
Filter and dry the reprecipitated components for identification.